

UNIT 5

QUESTION BANK

Short answer questions

1. A company wants to maximize profit under uncertain market conditions. Formulate a fuzzy objective function and constraints.
2. Describe the steps involved in designing a fuzzy controller for a temperature control system.
3. What is fuzzification and defuzzification in a fuzzy control system? Explain with examples.
4. Write down the general form of a fuzzy linear regression model and explain its components.
5. Describe one real-life application of fuzzy linear regression.
6. Explain how a rule base is used in a fuzzy logic controller.
7. Describe the role of inference engine in fuzzy logic controller design.
8. Discuss how a fuzzy controller handles non-linear systems effectively.
9. Define the concept of α -cut and apply it to convert a fuzzy constraint into a crisp interval.

Long answer questions

1. Discuss the concept of fuzzy constraints and fuzzy objective functions in optimization. How are they handled in fuzzy decision-making?
2. How is α -cut used in fuzzy optimization? Explain the role of α -cuts in converting a fuzzy problem into a crisp problem.
3. Discuss the steps involved in estimating parameters in fuzzy linear regression using the least squares method.
4. Compare fuzzy linear regression with interval regression and traditional regression models. Use diagrams or tables where appropriate.
5. Discuss the role of fuzzification and defuzzification in fuzzy control. How do they affect the performance of the controller?
6. Explain the basic structure of a fuzzy logic controller (FLC). Describe each component with the help of a neat block diagram.
7. Explain how fuzzy controllers handle uncertainty and imprecision in input data. Why is this useful in real-world control applications?